TCY Consulting Private Limited

TCY WorkHub — Employee & Daily Work Management Desktop Application

Updated Product & Technical Plan | Lean MVP | October 2026

> Updated scope: Supabase is now the primary backend platform for the MVP. Document management and video conferencing are explicitly removed from the current release. The desktop application will focus on employees, attendance, leave, projects, tasks, standups, meetings, notifications, dashboards, and reporting.

1. Product Vision

TCY WorkHub is an internal desktop workspace designed to replace manual attendance registers, Excel-based tracking, WhatsApp leave requests, scattered task updates, and disconnected daily-work processes with one simple company system.

The central UX concept is My Day: when an employee opens the application, they immediately see their attendance, today's meetings, standup, assigned work, deadlines, blockers, and important notifications.

The first release is intentionally small. It should become useful every working day before additional enterprise features are added.

2. Current MVP Scope

Module	MVP Scope
Authentication	Secure login, session management, role-based access.
Employee Management	Employee profile, department, designation, manager, joining date, employment status.
Attendance	Clock-in/out, daily history, monthly view, manager team attendance, admin correction.
Leave	Apply leave, approval/rejection, leave history, balances/status, notifications.
Projects	Create projects, members, owner, dates, status, priority, project overview.
Tasks	Assignment, status, priority, due date, comments, subtasks, Kanban/list views.
Daily Standup	Yesterday/today/blockers, standup history, manager overview, blocker tracking.
Standup Action Items	Convert a standup item into a task with owner, project, priority, and deadline.
Meetings	Internal meeting scheduling, participants, agenda, calendar, meeting status and notes.
Notifications	Leave decisions, assignments, deadlines, meetings, standup reminders.
Reports	Basic attendance, leave, project, task and workforce reports.

3. Explicitly Out of Scope for MVP

- Document management: no project file repository, document versioning, or file-sharing module in the first release.
- Video/audio conferencing: no built-in WebRTC conference system in the first release. Meetings are scheduling/coordination only.

- Meeting recording/transcription: deferred.
- Advanced AI: deferred until the core data and workflows are stable.
- Complex HR/payroll: outside the scope of this internal work-management system.

> This keeps the initial system substantially easier to build, secure, test, deploy, and maintain.

4. Users & Permissions

Role	Capabilities
Employee	Attendance, leave requests, own tasks/projects, standup, meetings, notifications, own profile.
Manager	Team attendance, leave approvals, project creation, task assignment, deadlines, team standups, reports.
Admin	Employee master data, departments, designations, manager mapping, attendance corrections, all projects, reports, configuration.
Super Admin (optional)	System-wide roles, permissions, security and platform configuration.

5. Updated Frontend Technology Stack

Technology	Use in TCY WorkHub	Recommendation
React + TypeScript + Vite	Application UI foundation	Keep
Tauri 2	Desktop packaging, Windows .exe, native capabilities	Keep
TanStack Query	API requests, server cache, loading/error states, refresh	Keep
React Hook Form + Zod	Forms and schema validation	Keep
Zustand	Lightweight client/UI state	Keep; do not store server data here
Tailwind CSS	Utility-first styling and design system	Add
shadcn/ui + Radix UI	Primary UI component system	Preferred over MUI
Lucide React	Consistent application icons	Add
AG Grid Community	Employee, attendance and admin data grids	Preferred for enterprise tables
FullCalendar	Leave, holidays, schedules, meetings and milestones	Keep
dnd-kit	Project Kanban drag/drop	Keep
Recharts	Attendance, leave, project and workforce charts	Keep
date-fns	Date/time calculations and formatting	Add
Sonner	Lightweight success/error/toast notifications	Add
cmdk	Desktop command palette / Ctrl+K search	Add
React Router	Application routing	Add/standardize

6. Why shadcn/ui Instead of MUI

For this application, shadcn/ui + Tailwind CSS + Radix UI is recommended over Material UI. The application is an internal productivity product rather than a public Material Design website, so a custom design system is more valuable.

- More control over spacing, typography, colors and component appearance.
- Works naturally with Tailwind and modern React/TypeScript applications.

- Easy to create a consistent TCY visual identity.
- Good primitives for dialogs, sheets, menus, tabs, popovers, command palettes and forms.
- Avoids excessive dependency on a large pre-styled component system.

> MUI is still a valid alternative if the team prioritizes ready-made enterprise components and wants less custom styling.

7. Recommended Desktop UI Design

The application should look and behave like a modern desktop productivity tool rather than a web page inside an .exe.

UI Area	Design Direction
Sidebar	Collapsible navigation: Home, My Work, Attendance, Leave, Projects, Calendar, Meetings, Team, Reports, Settings.
Top Bar	Global search/command palette, connection status, notifications, user menu.
Dashboard	Card-based My Day view with attendance, tasks, meetings, standup and alerts.
Tables	Dense but readable enterprise tables with search, filters, sorting, column control and pagination.
Forms	Short, grouped forms with inline validation and clear save/cancel actions.
Status	Consistent badges for task, leave, attendance and project states.
Kanban	Clear columns, compact cards, priority indicators and drag/drop.
Calendar	Month/week/day views with color-coded meetings, leave, holidays and milestones.
Notifications	Notification center plus non-blocking toast feedback.
Command Palette	Ctrl+K quick actions: search employee, create task, apply leave, open project, submit standup.
Responsive Desktop	Optimized for laptop/desktop screens; support compact and comfortable density modes later.

8. Suggested My Day Dashboard

Header: Good morning, [Employee] · connection status · notifications · profile

Top cards: Clock In / Working Time · Today's Tasks · Meetings · Leave Status

Main area: Today's Work list with task priority, project, deadline and status.

Side area: Standup submission, blockers, upcoming meetings and pending actions.

Manager version: Team attendance, pending leave approvals, overdue tasks, blockers and standup completion.

9. Data & Backend Architecture

For the MVP, use Supabase as the managed backend platform instead of deploying a separate Spring Boot server.

Recommended architecture:

Tauri 2 → React/TypeScript → TanStack Query → Supabase Auth / PostgreSQL / Realtime

This is particularly appropriate for a 5–7 employee internal application because it reduces infrastructure and deployment work while retaining PostgreSQL as the relational database.

Layer	Technology	Purpose
Desktop	Tauri 2	Windows desktop application and native capabilities.

Layer	Technology	Purpose
Frontend	React + TypeScript + Vite	Application UI.
Server state	TanStack Query	Fetch/cache/invalidate/refetch server data.
Authentication	Supabase Auth	Employee login and session handling.
Database	Supabase PostgreSQL	Employees, attendance, leave, projects, tasks, standups, meetings, notifications.
Realtime	Supabase Realtime	Live notifications and selected collaborative updates.
Validation	Zod	Client-side schema validation; database constraints remain authoritative.
Local settings	Tauri Store	Theme, UI preferences and lightweight desktop settings.
Offline queue	SQLite via Tauri plugin (optional)	Pending actions/drafts if offline support is needed.

10. Supabase Database Design

Table	Purpose
profiles / employees	Employee information and account/profile mapping.
departments	Company departments.
designations	Employee designations.
attendance	Daily clock-in/out and attendance status.
attendance_corrections	Admin changes with reason and audit information.
leave_types	Configured leave categories.
leave_requests	Employee leave applications and manager decisions.
projects	Project master data.
project_members	Employee-project membership.
tasks	Project work items, assignments, priority, status and deadlines.
task_comments	Task discussion/history.
standups	Daily/team standup sessions.
standup_responses	Yesterday/today/blockers responses.
meetings	Meeting schedules, agenda and metadata.
meeting_participants	Meeting attendance/participant mapping.
notifications	User notifications and read status.
audit_logs	Important administrative/security activity.
holidays	Company holiday calendar.

11. Supabase Security Model

The most important design rule is: Supabase Row Level Security (RLS) must be enabled and designed before production data is added.

- Employees can read/update only their permitted records.
- Employees can see their own attendance and leave data.
- Managers can access team data only where the manager relationship permits it.

- Admins can manage company-wide records.
- Project/task access should be restricted by project membership or role.
- Never place Supabase service-role keys inside the Tauri application.
- Use the publishable/anon client key from the desktop app with RLS enforcing access.
- Use database constraints and server-side policies as the ultimate authorization boundary.

12. Attendance Workflow

1. Employee opens My Day.
2. Employee clicks Clock In.
3. Supabase records the attendance event/date/time.
4. Dashboard changes to Working.
5. Employee clicks Clock Out at the end of the day.
6. Working duration is calculated/displayed.
7. Manager can view team attendance.
8. Admin can correct attendance with a reason; correction is audited.

13. Leave Workflow

1. Employee opens Leave → Apply Leave.
2. React Hook Form collects the request and Zod validates it.
3. Request is inserted into Supabase.
4. Manager receives a realtime/in-app notification.
5. Manager approves or rejects.
6. Employee sees the decision immediately.
7. Leave calendar/reporting reflects the approved status.

14. Task + Standup Workflow

1. Daily standup is generated for the team.
2. Employee submits yesterday, today and blockers.
3. Manager sees a consolidated team view.
4. Manager can convert an action item into a task.
5. Task receives project, owner, priority and deadline.
6. Employee updates task status from To Do → In Progress → Review → Done.
7. Kanban and dashboard views update through TanStack Query and selected realtime events.

15. Meetings

- Meeting scheduling is included in MVP.
- Meetings have title, agenda, participants, project/team association, start/end time and status.
- FullCalendar provides day/week/month views.
- Employees can see upcoming meetings from My Day.
- Meeting notes and action items can be captured.

- Actual audio/video conferencing is intentionally deferred.

16. State Management Strategy

Data Type	Tool
Employees/tasks/attendance fetched from backend	TanStack Query
API loading/error/cache/refetch	TanStack Query
Sidebar open/closed	Zustand
Theme/UI preferences	Zustand + Tauri Store
Selected filters/view preferences	Zustand where useful
Forms	React Hook Form
Validation schemas	Zod
Offline pending actions	SQLite optional

17. Frontend Project Structure

```
src/
├── app/
├── routes/
├── components/
│   ├── ui/
│   ├── layout/
│   └── shared/
├── features/
│   ├── auth/
│   ├── employees/
│   ├── attendance/
│   ├── leave/
│   ├── projects/
│   ├── tasks/
│   ├── standups/
│   ├── meetings/
│   ├── notifications/
│   └── reports/
├── lib/
├── hooks/
├── stores/
└── types/
```

18. MVP Development Phases

Phase	Deliverables
1. Foundation	Tauri 2, React/Vite/TS, Tailwind, shadcn, routing, Supabase project, environment setup.
2. Auth + Employees	Login, roles, profiles, departments, designations, manager hierarchy.
3. Attendance	Clock-in/out, history, monthly view, team view, corrections/audit.
4. Leave	Leave types, application, approval/rejection, calendar, notifications.
5. Projects + Tasks	Projects, members, task CRUD, Kanban, table/list, deadlines, comments.
6. Standups	Daily standup, blockers, history, action-item-to-task flow.

Phase	Deliverables
7. Meetings	Calendar, meeting scheduling, participants, agenda, notes.
8. Reports + Polish	Dashboards, exports, UX refinement, security/RLS review, testing, installer.

19. Reports & Dashboards

- Attendance percentage by employee/team/month.
- Late and absent counts.
- Leave requested/approved/rejected.
- Open/completed/overdue tasks.
- Project progress and project status.
- Standup submission rate.
- Blockers by team/project.
- Meeting schedule and attendance metadata.

20. Desktop-Specific Features

- Windows .exe installer through Tauri.
- Desktop notifications for meetings, leave decisions, task assignments and reminders.
- Tray behavior can be added for reminders.
- Connection indicator: Online / Reconnecting / Offline.
- Automatic update strategy after the first stable release.
- Secure Tauri IPC; avoid exposing privileged APIs directly to the frontend.
- Local preferences through Tauri Store.
- Optional SQLite offline queue only if offline workflows become a real requirement.

21. Security Requirements

- Supabase RLS for all company tables.
- Never expose service-role credentials in the desktop application.
- Use strong authentication and session handling.
- Validate all user input with Zod plus database constraints.
- Audit administrative attendance corrections and permission-sensitive actions.
- Use least-privilege database policies.
- Protect environment/configuration secrets.
- Use HTTPS/TLS for production connections.
- Periodically review RLS policies using test accounts for Employee, Manager and Admin.

22. Offline Strategy

Offline support should be deliberately limited in the first version. The desktop app can continue to display some cached data through TanStack Query, but authoritative writes should normally require connectivity.

If offline actions become necessary, use SQLite as a small queue:

```
User action → SQLite pending queue → connection returns → sync → Supabase → mark synced
```

Do not use SQLite as the primary company database.

23. Future Features — Not MVP

- Project document management.
- Audio/video conferencing with WebRTC.
- Meeting recording and transcription.
- AI standup summaries.
- AI meeting summaries and action extraction.
- Natural-language workforce/project reporting.
- Advanced analytics and forecasting.
- Mobile application.
- SSO integration.
- Advanced HR/payroll integrations.

24. Recommended Final Technology Stack

Layer	Selected Technology
Desktop runtime	Tauri 2
Frontend	React + TypeScript + Vite
Styling	Tailwind CSS
UI components	shadcn/ui + Radix UI
Icons	Lucide React
Server state	TanStack Query
Client state	Zustand
Forms	React Hook Form
Validation	Zod
Routing	React Router
Tables	AG Grid Community
Calendar	FullCalendar
Kanban	dnd-kit
Charts	Recharts
Dates	date-fns
Notifications	Sonner + Tauri desktop notifications
Command search	cmdk
Backend platform	Supabase
Authentication	Supabase Auth
Database	Supabase PostgreSQL
Realtime	Supabase Realtime
Local settings	Tauri Store
Offline queue	SQLite via Tauri plugin, optional

25. Final Architecture

```
TCY WorkHub Desktop

Tauri 2
↓
React + TypeScript + Vite
↓
TanStack Query + Zustand
↓
shadcn/ui + Tailwind + Radix + Lucide
↓
Supabase Auth + PostgreSQL + Realtime
↓
Optional Tauri Store / SQLite for local preferences and offline queue
```

26. Final Recommendation

For a 5–7 employee internal system, this is a much better first architecture than introducing a separate Spring Boot server, Redis, object storage, WebRTC infrastructure, and multiple services from day one.

The MVP should concentrate on the workflows employees will use every day: clock in → see My Day → submit standup → work on tasks → attend meetings → update work → clock out. Managers should have a corresponding team view, while admins maintain employee and company configuration.

> Once TCY is actively using the application, real usage will tell you what deserves the next investment. Document management, conference calls and AI features can then be added without bloating the first release.

Document status: Updated according to the requested MVP scope and technology/UI direction.